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Leakage-Safe Taxi-Out Time Prediction at Ten European Airports

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Abstract

Taxi-out is a major source of avoidable delay, fuel burn and emissions at busy airports, and its duration is hard to predict at the moment an aircraft leaves the stand. We predict taxi-out time for departures from ten major European airports, using the 2,085,047 departures from 2025 released for the PRC Data Challenge 2026 (team trustworthy-zipper) and scoring on four months of 2026. Every feature is built only from information available at off-block: recent departure and arrival traffic, departure delays, weather reports, stand-to-runway distance from open map data, and a leave-one-month-out typical taxi time for each airport, stand and runway. A gradient boosted regression tree model reaches a root mean square error (RMSE) of 308.6 s on four held-out months, 24% below an airport-and-runway median baseline (403.7 s), and performs as well in February and June as on the originally scored January and July. The typical taxi time alone carries 44% of the model's split gain. We show that the take-off time present in the ranking data reconstructs the target and exclude it, although it would lower the measured error by about 40 s. Snow is the largest remaining error source: snowy departures have an RMSE of 707 s against 306 s otherwise. On the official January and July 2026 test set the model scored an RMSE of 571.8 s, 10% better than the typical taxi time used on its own (634.7 s).

Keywords: taxi-out time prediction; gradient boosting; feature importance; data leakage; airport operations

Abbreviations: AOBT: Off-block time from the flight plan (AOBT_3), RMSE: Root mean square error

1. Introduction

Taxi-out time is the time from off-block at the stand to take-off. At congested airports it is far from constant: queues at the runway, runway configuration, de-icing and the distance from stand to runway can each add several minutes. EUROCONTROL tracks the excess over an unimpeded value as a key performance indicator, because extra taxi time burns fuel, adds emissions and noise, and signals that runway capacity is limiting operations [1]. Accurate predictions made at off-block support departure sequencing, fuel planning and stand allocation.

The Performance Review Commission (PRC), with the OpenSky Network, runs an annual open data challenge with the aim of developing methods for the wider aviation community from open data, with code and documentation shared publicly [2]. Earlier editions addressed take-off weight and fuel burn estimation [3, 4]. The 2026 edition asked teams to predict taxi-out time for departures from ten European airports. Teams trained on all 2025 movements and were ranked by RMSE on January and July 2026, with February and June 2026 added in a final phase to test generalisation beyond the originally scored months.

Taxi-time prediction has been studied with queuing models, statistical regression, fuzzy rule-based systems and machine learning [5, 6, 7, 8, 9, 10, 11], usually at one or a few airports and with data from the airport operator. The challenge setting differs in two ways: the data covers ten large airports at

once, and some fields present in the ranking data are not known at off-block. This makes leakage control a central design question.

The contributions of this paper are:

1. A leakage-safe feature set for ten airports, built only from information known at off-block and from open external data on weather, runways and stands.
2. A leave-one-month-out typical and unimpeded taxi time per airport, stand and runway, which is the single most important predictor.
3. An explicit leakage analysis showing that take-off time minus planned off-block time reconstructs the target, with the measured cost of excluding it.
4. Evidence that the model generalises across months and an analysis of where it fails, chiefly in snow.

The rest of the paper is organised as follows. Section 2 reviews previous work. Section 3 describes the data and features. Section 4 presents the model, performance metric, feature importance procedure and leakage controls. Sections 5 and 6 report results on model performance and feature importance. Section 7 concludes and sets out future work.

2. Literature review

Previous work on taxi-time prediction falls into four method families. We summarise each briefly and close with the gaps this paper addresses.

Queuing models. Early work treated the departure runway as a server and taxi-out time as travel time plus queueing delay. Idris et al. [5] showed that the number of departures taking off while an aircraft taxis is the strongest single explanatory variable for taxi-out time. Simaiakis and Balakrishnan [6] built a queuing model of the whole departure process and used it to predict taxi-out times and runway throughput. These models explain congestion well but need careful calibration per airport and runway configuration.

Statistical regression. Ravizza et al. [7] combined a multiple linear regression on features such as taxi distance, number of turns and concurrent traffic with a ground movement model, and showed clear gains over the fixed taxi times then used in planning. Regression models are transparent but capture interactions between features only if these are specified by hand.

Fuzzy rule-based systems. Chen et al. [8] used fuzzy rule-based systems to model the uncertain, non-linear relation between airport conditions and taxi time. Ravizza et al. [9] compared regression, fuzzy rule-based and other approaches on the same data and found that the choice of features mattered at least as much as the choice of method.

Machine learning. Lee et al. [10] compared several machine learning methods with fast-time simulation for taxi-out prediction at a large US hub. Wang et al. [11] compared neural networks, linear and polynomial regression, gradient boosted trees and random forests across several airports, and found that a small subset of features gives nearly the same accuracy as the full set.

Research gaps. Most studies use one or a few airports and operator data with precise times and taxi routes. Few discuss which fields would really be known at the moment of prediction, so leakage is rarely analysed. Weather is often reduced to coarse indicators, and generalisation to unseen months or seasons is seldom tested. This paper addresses these gaps with a ten-airport open dataset, a feature set restricted to off-block information, an explicit leakage analysis and a month-level generalisation test.

3. Data

The challenge data [2] covers all movements at ten European airports: Amsterdam (EHAM), Barcelona (LEBL), Frankfurt (EDDF), Istanbul (LTFM), London Heathrow (EGLL), Madrid (LEMD), Munich (EDDM), Paris Charles de Gaulle (LFPG), Rome Fiumicino (LIRF) and Zurich (LSZH). Each record carries the airport, runway, stand, aircraft type and wake category, operator, scheduled time, planned off-block time (AOBT), the movement time and, for training only, the block time and the target taxi-out time. The target equals movement time minus block time exactly in all training rows. Table 1 lists the sets used.

Table 1. Training and ranking sets

Set	Period	Departures
Training	Jan–Dec 2025	2,085,047
Ranking	Jan 2026	152,724
Ranking	Feb 2026	144,149
Ranking	Jun 2026	181,789
Ranking	Jul 2026	192,118

We add three open sources: hourly and special METAR reports from the Iowa Environmental Mesonet [12], runway threshold coordinates from OurAirports [13], and stand positions from OpenStreetMap [14], downloaded on 30 September 2026. Suspicious rows are flagged rather than deleted during cleaning. For training we keep taxi-out times between 60 s and 7,200 s; the ranking set is not filtered. The organisers' final release also revised arrival times for the original January and July flights by up to ±12 s, which changes the 30-minute arrival count by one for about 23% of those flights.

All features are computed relative to an anchor time known at off-block: the AOBT, or the scheduled time when the AOBT is missing (about 1–1.5% of flights). Block time is never used, because it is blank in the ranking data.

3.1 Aircraft and airport operational features

Table 2. Aircraft and airport operational features

Feature	Type	Description
ADEP	Categorical	Departure airport
RUNWAY	Categorical	Departure runway, prefixed with the airport
AIRCRAFT_TYPE	Categorical	ICAO aircraft type
AIRCRAFT_OPERATOR	Categorical	Operating airline
aircraft_size	Categorical	Wake category L, M, H or J; UNK if missing
pushback_hour	Numerical	UTC hour of the anchor time
is_weekend	Binary	Saturday or Sunday
month	Numerical	Month of the anchor time; final model only
dep_delay_min	Numerical	AOBT minus schedule, minutes, clipped to −120 to 600
aobt_missing	Binary	AOBT missing, anchor fell back to schedule

Table 3. Airport congestion features

Feature	Type	Description
planes_pushed_last_15min	Numerical	Departures at the airport in the 15 min before the anchor
planes_pushed_last_30min	Numerical	As above, 30 min
planes_pushed_last_60min	Numerical	As above, 60 min
arrivals_last_30min	Numerical	Landings at the airport in the 30 min before the anchor

3.2 Airport congestion features

Counts include only events strictly before the anchor, never the same minute or the flight itself. They are built over the whole year at once, so the first hour of each month is not undercounted, and were checked by hand on 300 random flights with no mismatches.

3.3 Stand and runway features

Table 4. Stand and runway features

Feature	Type	Description
stand_area	Categorical	First character of the stand code; groups under 1% merged into OTHER
typical_taxi_sec	Numerical	Median taxi-out time for airport + stand + runway
unimpeded_taxi_sec	Numerical	10th percentile of the same
stand_runway_km	Numerical	Straight-line distance from stand to runway threshold

The typical and unimpeded times use cells with at least 10 flights and fall back to airport + runway, then airport. Training rows use leave-one-month-out statistics, so a January flight's value comes only from the other eleven months; ranking rows use all of 2025. Stand codes are normalised and matched to OpenStreetMap parking positions; matches more than 8 km from the runway are discarded. Coverage is 96–99% at eight airports, 90% at LEBL and 57% at EDDM, and within each airport longer distance goes with longer taxi time (correlation 0.16–0.56).

3.4 Weather features

Table 5. Weather features

Feature	Type	Description
visibility_miles	Numerical	Visibility, statute miles
wind_kt	Numerical	Wind speed, knots
temp_c	Numerical	Air temperature, °C
is_fog	Binary	Fog reported
is_snow_ice	Binary	Snow or freezing conditions reported

Each flight takes the latest report at or before its anchor time. Duplicate reports are removed and a missing value takes the previous report's value. Precipitation amounts are left out because they are

not reported reliably outside the United States.

4. Prediction model and feature importance

4.1 Model

We use gradient boosted regression trees as implemented in XGBoost [15]. The model is an additive ensemble of K regression trees, each fitted to the residual errors of those before it. Training minimises a squared-error loss plus a penalty on tree complexity, as shown in Equation 1.

$$\mathcal{L} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{k=1}^{K} \left(\gamma T_k + \frac{1}{2} \lambda \| w_k \|^2 \right) \tag{1}$$

Here T_k is the number of leaves of tree k, w_k its leaf weights, and γ and λ are regularisation parameters. Squared error matches the ranking metric directly. Trees use histogram-based splits and native handling of categorical features, so airport, runway, aircraft type and operator need no one-hot encoding. Gradient boosted trees were chosen because they handle mixed feature types and missing values without preprocessing, and they performed among the best methods in comparable work [11].

4.2 Performance metric

Root mean square error. The challenge ranks teams by RMSE in seconds over all ranking departures, as shown in Equation 2.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{2}$$

Because errors are squared, a few very long taxi-outs, for example during de-icing, weigh heavily on the score.

Baseline. As a reference we predict the median taxi-out time of each airport and runway combination in the training data.

4.3 Feature importance identification procedure

We measure importance in two ways. First, an ablation adds feature groups one at a time and records the change in held-out RMSE; this test used LightGBM for speed. Second, for the XGBoost model evaluated in Section 5 we report each feature's share of total split gain, the summed loss reduction from all splits on that feature.

4.4 Leakage control

The ranking data contains the actual take-off time, even though it marks the end of the period being predicted. Because the target is take-off time minus block time, and the planned off-block time is close to block time, take-off time minus AOBT is a noisy copy of the target; it correlates with the target at $r = 0.64$. Adding it cut held-out RMSE from 315 s to 276 s in the ablation test, more than any legitimate feature. A system working at real off-block could not know the flight's take-off time, so we exclude it. We also avoid monthly aggregates for 2026, such as EUROCONTROL taxi-out indicators [1], because they summarise the ranking-set taxi times.

5. Results: Model configuration and tuning

5.1 Experimental setup

We hold out January and July 2025, the months that mirror the original ranking set, as the test set. The other ten months are split randomly 90/10 into training and validation; the validation set drives early stopping and tuning. The month feature is withheld during this evaluation because the test months are absent from training.

Hyperparameters were tuned with Optuna [16] (TPE sampler, 10 trials) on a 400,000-row sample at learning rate 0.1. The final settings are listed in Table 6.

Table 6. XGBoost hyperparameters

Parameter	Value
max_depth	9
min_child_weight	1.9
subsample	0.72
colsample_bytree	0.68
reg_lambda (λ)	1.7
learning_rate	0.05
Number of trees	1,840

The production model is refitted on all 2025 departures with the month feature added, using 1.1 times the best early-stopping iteration (1,673 to 1,840 trees). For the final phase we also trained a diagnostic model with the same settings that holds out all four scored months of 2025 at once; it is used only for evaluation.

5.2 Baseline

The airport-and-runway median baseline gives an RMSE of 409.1 s on held-out January and July 2025. Predicting a single overall mean gives 452 s.

5.3 Performance

The tuned model scores 310.7 s on held-out January and July 2025, 24% below the baseline. Validation RMSE on the random split is 253.7 s; the gap shows that a random split overstates performance, because flights from the same day appear on both sides. Table 7 shows the generalisation test.

Table 7. Generalisation test with all four scored months held out

Held-out months (2025)	Model RMSE (s)	Baseline RMSE (s)
Jan + Jul	312.9	408.5
Feb + Jun	304.0	398.6
All four	308.6	403.7

The model does at least as well on February and June as on the originally scored months, so it has not been overfitted to January and July. By month, RMSE is 308.3 s in January, 343.1 s in February, 269.4 s in June and 316.5 s in July. Holding out two extra months moved the January and July score only from 310.7 s to 312.9 s.

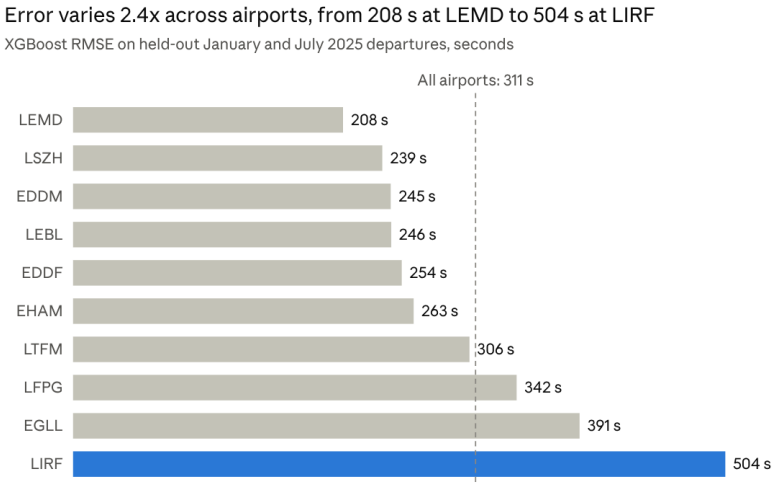


Figure 1. RMSE per airport on held-out January and July 2025 departures

Error differs strongly between airports (Figure 1). Rome Fiumicino and London Heathrow have the highest errors, while the Spanish, Swiss and German hubs sit between 208 and 254 s. This is probably linked to runway configuration changes and stand matching quality at those airports, which our features capture only partly.

5.4 Official challenge result

The official scorer evaluates January and July 2026 only. Our first upload, which also covered February and June (670,790 flights), was rejected because the scorer's truth set holds the 344,841 January and July departures. Restricted to those flights, the same predictions were accepted (Table 8).

Table 8. Official results on the January and July 2026 test set

Submission	Scored flights	Official RMSE (s)
XGBoost model (v2)	344,841	571.8
Typical taxi time only (v3)	344,841	634.7

The official RMSE of 571.8 s is far above the 310.7 s measured on held-out January and July 2025. The typical taxi time alone rises in the same way, from about 409 s for the comparable 2025 baseline to 634.7 s, so the gap comes from the 2026 test data rather than a fault in the model. The most likely cause is the target filter: our 2025 evaluation used only taxi-out times between 60 s and 7,200 s, while the official truth set includes every departure, and a few very long values dominate a squared-error metric. The higher snow share of January 2026 adds to this. The model still improves on the typical taxi time by 10% on the official data.

At the time of submission the leading teams scored between 208 s and 230 s. We have not seen their methods; a gap of this size is consistent with using the take-off time present in the ranking data, which we excluded as leakage (Section 4.4).

6. Results: Feature importance

6.1 Performance of different feature groups

Each feature group lowers held-out RMSE (Table 9). The largest step comes from the basic time, traffic and weather features together with the raw airport, runway and aircraft type (452 s to 339 s). The typical taxi time, delay and arrival features then remove a further 23 s. Stand-to-runway distance adds little on top, because the typical taxi time already encodes each stand and runway pair, but it helps stands with too few flights for a reliable typical value.

Table 9. RMSE as feature groups are added (LightGBM, January and July 2025 held out)

Features	RMSE (s)
Predict the mean	452
Time, traffic, weather + raw airport, runway, aircraft type	339
+ typical and unimpeded taxi time, delay, arrivals	315.9
+ stand-to-runway distance	315.4
+ take-off minus AOBT (excluded as leakage)	276

6.2 Feature importance identification

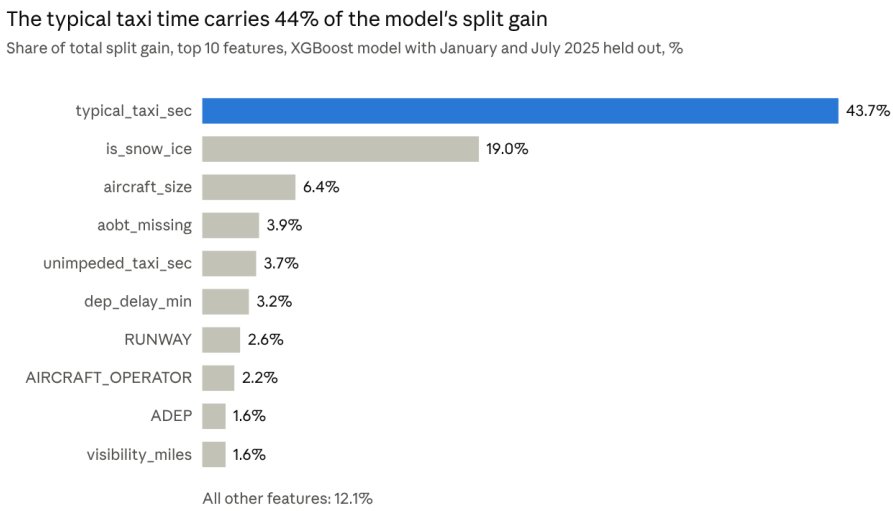


Figure 2. Feature importance by share of split gain

The typical taxi time dominates the ranking (Figure 2). It summarises the taxi route, the usual runway queue and airport-specific procedures in one number, which is consistent with earlier findings that a small subset of features carries most of the predictive power [11]. The high rank of the snow flag reflects a few very large errors rather than many small ones: it applies to only 0.5% of training departures. Aircraft size and departure delay follow; late departures tend to meet different runway queues from those planned.

6.3 Weather and snow

Snowy departures in the held-out months (2,482 flights, 0.7%) have an RMSE of 706.7 s, against 305.9 s for the rest. This matters for the 2026 score because the weather mix shifted in January only (Table 10).

Table 10. Share of departures with snow or fog reported

Month	Snow 2025	Snow 2026	Fog 2025	Fog 2026
Jan	1.6%	4.6%	3.9%	3.2%
Feb	3.0%	1.9%	2.0%	2.9%
Jun	0.0%	0.0%	0.2%	0.1%
Jul	0.0%	0.0%	0.3%	0.1%

January 2026 is therefore expected to score worse than its 2025 counterpart, while February, June and July show no adverse shift.

7. Conclusions

This paper predicted taxi-out time for departures from ten European airports using only information available at off-block. A gradient boosted tree model on traffic, delay, weather, stand and runway features reduced RMSE by about 24% against an airport-and-runway median, from 403.7 s to 308.6 s across four held-out months. The leave-one-month-out typical taxi time per airport, stand and runway was by far the most important feature, followed by snow, aircraft size and departure delay. The model generalised to February and June as well as to the originally scored January and July. We deliberately excluded a take-off-time feature that would have lowered the measured error by reconstructing the target, and recommend that future challenges blank this field in ranking data. On the official January and July 2026 test set the model scored 571.8 s, 10% better than the typical taxi time alone (634.7 s); the gap to our 2025 estimate mainly reflects extreme taxi-out values that our training filter excluded.

Several directions could improve the model further:

1. **Snow and de-icing.** January 2026 has about three times the snow share of January 2025 (4.6% against 1.6% of departures), and snowy departures carry more than double the error. Features describing snow or freezing precipitation over the preceding hours, and airport de-icing procedures, are the clearest next step.
2. **Grouped validation.** Tuning on a validation set grouped by day or month, instead of a random split, would give a fairer target and may favour more regularised trees.
3. **The month feature and the target cap.** The month feature was added only to the final model, and training excludes taxi-out times above 2 h. Both choices should be tested directly, especially for long snowy taxi-outs.
4. **Runway configuration.** Live runway configuration data, for example published operating directions at Frankfurt or Amsterdam, could reduce the large errors at airports such as LIRF and EGLL.
5. **Train–serve consistency.** Training rows see typical times from eleven months, ranking rows from twelve; measuring this small difference would complete the leakage analysis.

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Author contributions

- Kabir Ansari: Project Administration, Software, Writing (Original Draft). GitHub repository, requirements and documentation, airport domain research, and drafting this paper.
- Maham Baber: Formal Analysis, Software. Supporting the modelling and evaluation with Oscar Denche.
- Oscar Denche: Methodology, Formal Analysis, Software, Validation. Validation strategy, gradient boosting models, feature importance analysis, final-phase submission and generalisation checks.
- Anahita Hessami: Investigation, Data Curation. Exploratory data analysis, profiling of the raw training data, and analysis of anomalies and outliers in the taxi-out target.
- Harsh Narendra: Data Curation, Methodology, Software. External weather, runway and stand data, and the leakage-safe feature enrichment.
- Kamran Rzayev: Data Curation, Software. Data cleaning pipeline, quality flags and the movement–flight join.
- Elisa Schezzini: Formal Analysis, Software. Hyperparameter tuning with Optuna and comparison of global and airport-specific models.
- Mihai-Catalin Vasile: Software, Validation. Backend pipeline integrating the cleaned and enriched data into the training matrix, and leakage checks.

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Open data statement

The PRC Data Challenge 2026 dataset of airport movements is provided by the EUROCONTROL Performance Review Commission and the OpenSky Network and is available to registered teams through the challenge [2]. All external data are openly available: METAR weather reports from the Iowa Environmental Mesonet [12], runway coordinates from OurAirports [13] and stand positions from OpenStreetMap under the ODbL licence [14]. Cached copies of the external data used in this paper (weather_cache.parquet, runways_10_airports.csv, osm_stands.csv) are included in the code repository.

Reproducibility statement

All source code is public under the GNU GPLv3 at https://github.com/kabirans7/PRC_Data_Challenge_2026. The repository README gives the run order: enrichment.py builds the features from the challenge files and the cached external data, and the notebook PRC_Eurocontrol_ML_Pipeline_adapted.ipynb trains, evaluates and tunes the model, writes the predictions, and produces the numbers in the tables and the data behind the figures. The code uses Python with pandas, NumPy, PyArrow, XGBoost, scikit-learn, Optuna and joblib; exact versions are pinned in requirements.txt. Training runs on a CPU, with an optional GPU setting.

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